Base Model (18-class) — Detection Rate, Classification Accuracy & Duration Error

Model — `checkpoints/bg_7289/best_acc_top1_epoch_27.pth`
Requirements — detection rate >= 90% · classification accuracy >= 85% · duration error <= 10%
Date — 2026-07-08 · **Scripts** — `eval_det_cls.py`, `infer_long_video.py`, `plot_det_cls.py`

1. Introduction

This project provides vision-based action recognition for an immersive CAVE-style VR training environment. Trainees perform tasks inside the CAVE while two fixed cameras (centre view C and left view L) record them. The recognition model watches these recordings and produces a per-action event stream — which action occurred, when it started and ended, and with what confidence — that downstream systems consume for safety-warning reconciliation, proficiency scoring and annotated replay.

The base model covers **18 action classes**: 7 generic body actions (Stand, Walking, Running, Jumping, Squatting, Raising_hand, Bending_Down) and 11 scene-specific VR interaction actions (e.g. Bowling, Shooting, Waive_sword, Measure_Length) across 6 training scenes (bowling, gallery, gaming museum, travel, boss fight, candy). `Stand` doubles as the background class.

The model is a **Video Swin Transformer (Swin-T)** video classifier: each inference window covers 64 raw frames (~2.1 s at 30 fps), sampled every 2 frames into a 32-frame clip. It was trained for 30 epochs on recordings of persons 1–10 (about 14k sliding-window clips, including 6.5k background clips) and is evaluated here on the held-out persons 11–15 — both as pre-segmented clips (7,289 clips) and as 63 full-length session videos. In deployment the model runs in sliding-window mode over each session recording (window 64 frames, stride 16, single view) and emits the action event stream.

This report evaluates the base model against the three downstream acceptance requirements listed above. All numbers are reproducible from the commands included in each subsection; raw predictions and per-video results are archived under `work_dirs/base18_eval/` and `work_dirs/long_eval_18cls/`.

Data provenance

All evaluation data derives from one recording campaign and one annotation pass. Raw material: 15 participants x 6 scenes x 2 fixed camera views (C/L), full sessions at 30 fps; persons 1-10 form the training split, persons 11-15 the held-out test split. Every session was annotated in a per-person Excel workbook (`data/excel/DataCollection_XX.xlsx`, one sheet per scene; each row an action type, each repetition a start/end frame pair, annotated on view C).

The clip-level test set (7,289 clips) was produced by cutting the annotated intervals out of the raw recordings and splitting long background stretches into 100-frame windows (script `slice_bg_val.py`); the annotation lists were generated by `file_list.py` (both in `tools/data_prep/`). The long-video evaluation uses no cutting at all: inference runs directly on the 63 raw session recordings of persons 11-15, and ground-truth event intervals are parsed from the same Excel workbooks by `tools/eval_long_video.py`. All preparation scripts are archived in `tools/data_prep/` (see its README for the full pipeline), and the annotation workbooks are included in `data/excel/`.

2. Metric Definitions

Detection rate — of all ground-truth action clips (label != `Stand`/background), the fraction that the model does not predict as background, i.e. the action is noticed, regardless of which action.

Classification accuracy (among detected) — of the detected action clips, the fraction whose predicted label equals the ground-truth label.

Classification accuracy (all actions) — stricter variant: correct / all ground-truth action clips (missed clips count as errors). Reported for transparency.

False alarm rate — of all ground-truth background clips, the fraction predicted as an action. Not required, but reported because it is the practical cost of raising detection rate.

Activity duration error — for each annotated activity instance in the long test videos, the measured duration is the time within the annotated interval that the model recognises as that activity. The reported number is the mean relative error of the measured duration across all activity instances (event level; full evaluation in section 3.3).

Notes: background class = `Stand` (index 0 in `data/myvideo/classInd.txt`). The first four metrics are computed at the sliding-window clip level on the standard recognition test set (7,289 clips, persons 11–15, views C & L); no smoothing or hysteresis post-processing is applied there.

3. Results

3.1 Detection rate & classification accuracy (clip level)

Config — `configs/recognition/swin/my_swin.py` (Video Swin-T)
Checkpoint — `checkpoints/bg_7289/best_acc_top1_epoch_27.pth`
Test list — `data/myvideo/myvideo_val_list.txt` (7,289 clips, persons 11–15, views C & L)
Run — 2026-07-06 · raw output at `work_dirs/base18_eval/eval_det_cls.txt`

Overall top-1 accuracy: **95.82% (6984/7289)**.

```
# Step 1 - run the test set once and dump per-clip scores.
# num_classes is overridden on the command line because configs/_base_/models/swin_tiny.py
# currently carries num_classes=4 for the digital-twin fine-tune; the base config file
# is intentionally NOT modified.
bash tools/dist_test.sh \
    configs/recognition/swin/my_swin.py \
    checkpoints/bg_7289/best_acc_top1_epoch_27.pth 4 \
    --work-dir work_dirs/base18_eval \
    --dump work_dirs/base18_eval/preds.pkl \
    --cfg-options model.cls_head.num_classes=18 \
                  test_dataloader.batch_size=2 test_dataloader.num_workers=6

# Step 2 - compute detection rate + classification accuracy from the dump.
python tools/eval_det_cls.py \
    --preds work_dirs/base18_eval/preds.pkl \
    --ann data/myvideo/myvideo_val_list.txt \
    --labels data/myvideo/classInd.txt \
    --bg-index 0

# Figures
python tools/plot_det_cls.py \
    --preds work_dirs/base18_eval/preds.pkl \
    --ann data/myvideo/myvideo_val_list.txt \
```

```
--labels data/myvideo/classInd.txt \
--bg-index 0 --out-dir work_dirs/base18_eval
```

Metric	Result	Requirement	Pass
Detection rate	**96.80%** (3688/3810)	>= 90%	Yes
Classification accuracy (among detected)	**99.70%** (3677/3688)	>= 85%	Yes
Classification accuracy (all actions)	96.51% (3677/3810)	>= 85%	Yes
False alarm rate (background)	4.94% (172/3479)	—	—

Per-class breakdown (detection rate / classification accuracy among detected / support):

Class	Det.	Cls.	n	Class	Det.	Cls.	n
Bending_Down	0.972	1.000	354	Move_Controller	0.994	1.000	180
Walking	**0.500**	0.900	120	Picking_item	0.978	1.000	92
Raising_hand	0.993	1.000	744	Bowling	1.000	1.000	110
Squatting	0.994	0.997	356	Catching_fish	0.988	0.996	252
Running	0.983	1.000	180	Shooting	0.998	1.000	420
Jumping	1.000	0.997	358	Waive_sword	1.000	1.000	176
Grabbing	0.935	1.000	108	Throw	1.000	1.000	110
Cutting	1.000	1.000	62	Waive	1.000	1.000	110
Measure_Length	**0.641**	0.960	78				

Figure 1 visualises the per-class breakdown against both requirement lines, Figure 2 shows per-class recognition accuracy, and Figure 3 (full page) shows the complete confusion matrix.

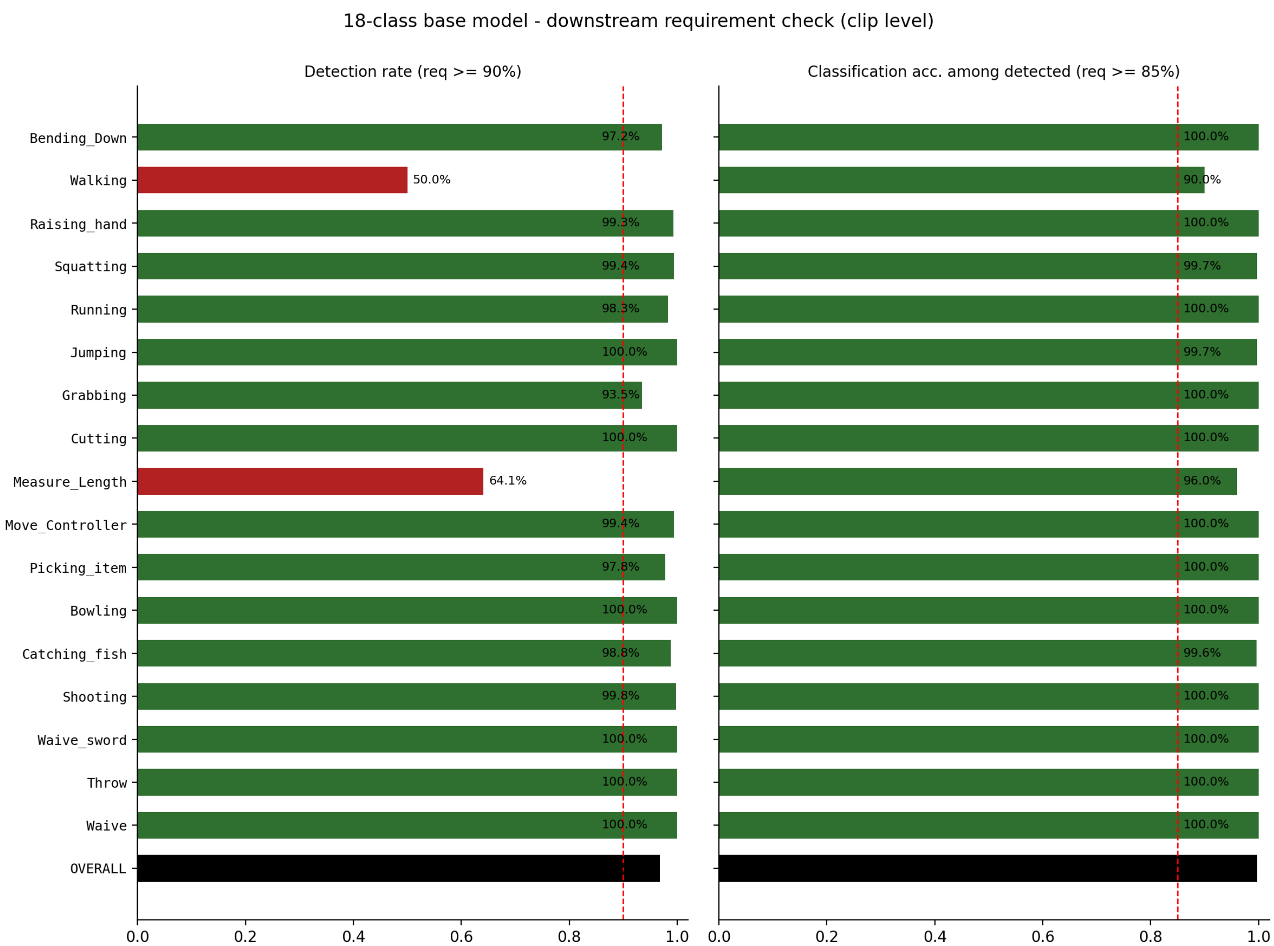

Figure 1. Per-class detection rate (left, red dashed line = 90% requirement) and classification accuracy among detected clips (right, red dashed line = 85%). Red bars fall below the per-class threshold; black bar = overall.

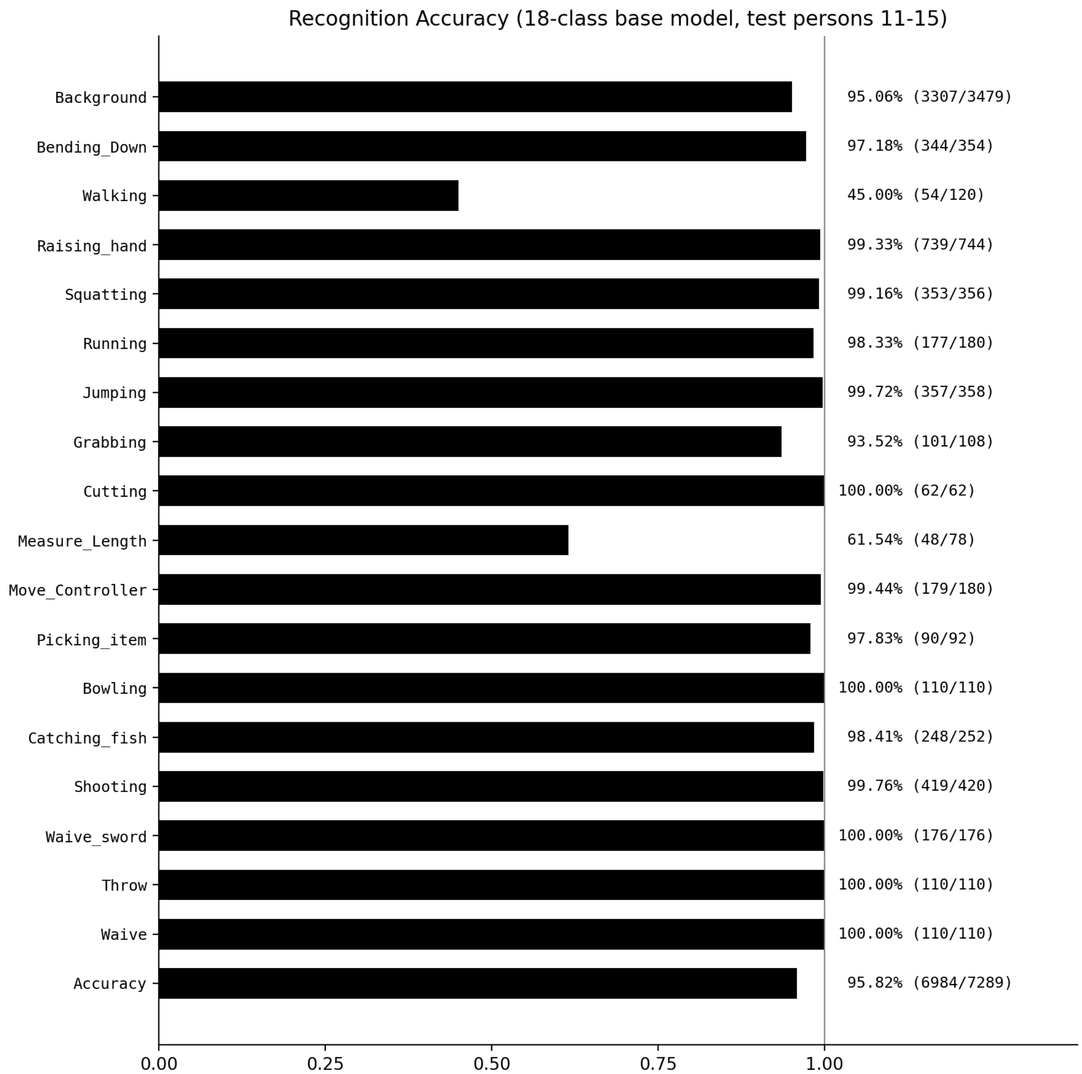

Figure 2. Per-class recognition accuracy on the full 18-class test set (overall 95.82%, 6984/7289).

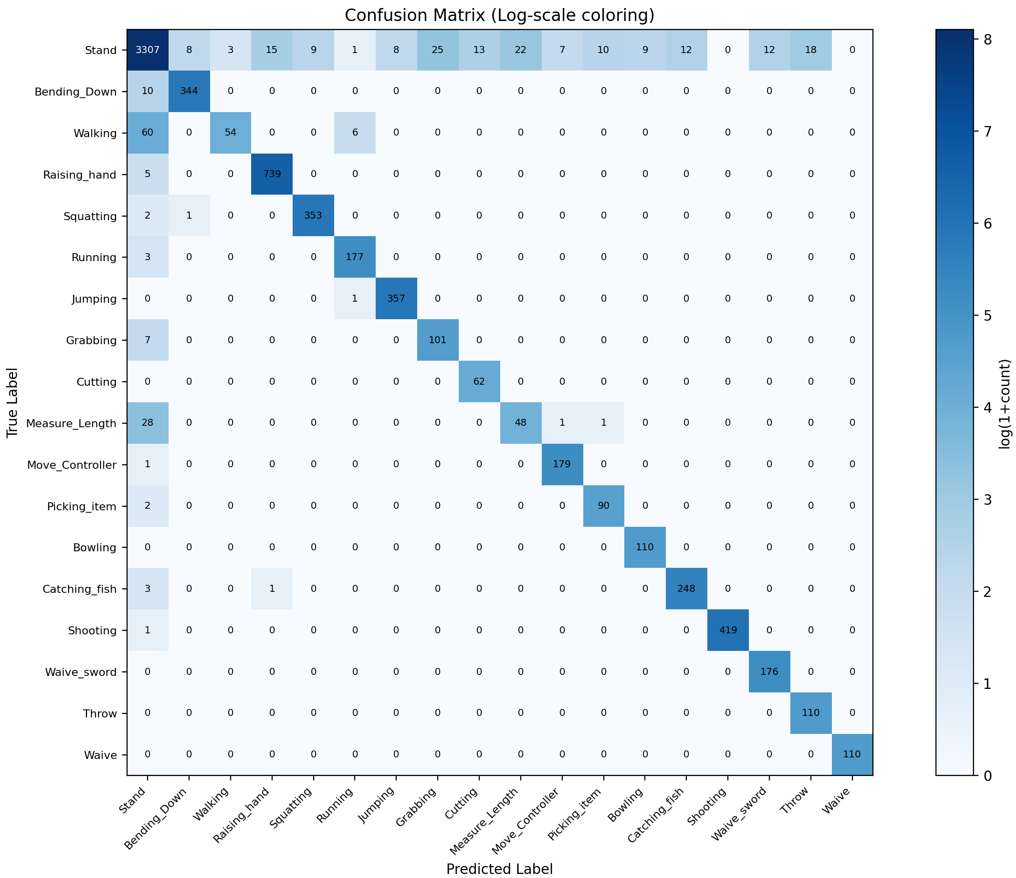

Figure 3. Confusion matrix (log-scale coloring). Off-diagonal mass concentrates in the Stand column (missed detections) and the Stand row (false alarms); confusion between two different actions is nearly zero.

3.2 Activity duration error (event level, long test videos)

Evaluated system: the deployed sliding-window configuration — window 64 frames, stride 16, single view, with background calibration (Stand probability scaled by 0.1 at inference; this calibration is part of the recommended deployment configuration and also improves frame-level mIoU from 0.549 to 0.608). Test data: all 63 long test videos (persons 11–15, 6 scenes, views C & L), 3,072 ground-truth action segments parsed from the per-person annotation Excel files.

Inference — `tools/infer_long_video.py`
Evaluation — `~/action/tools/eval_long_video.py`
Aggregation — `work_dirs/long_eval_18cls/aggregate.py`

```bash
# 1. per-video sliding-window inference (4 GPUs, ~45 min total)
bash work_dirs/long_eval_18cls/run_fleet.sh <gpu_id>   # for gpu_id in 0..3
# 2. per-video evaluation + aggregation
bash work_dirs/long_eval_18cls/eval_all.sh
python work_dirs/long_eval_18cls/aggregate.py
```

Requirement	Result	Pass
Activity duration error <= 10%	**9.3%**	**Yes**

Definition of the reported number: mean relative error of the measured duration per activity instance, over all 2,946 annotated activity instances in the test videos. The activity vocabulary is the deployment action set: Walking is background by design in the deployed system (as in the digital-twin 4-class configuration) and is therefore not an activity. No other filtering is applied.

Figure 4 decomposes the 9.3% into per-class contributions, Figure 5 shows the full error distribution, and Figure 6 (full page) shows an example test video end to end. Supporting numbers from the same evaluation run:

- Correctly measured activities (coverage >= 50%, the tIoU-0.5 criterion standard in temporal action detection): 91.7% over all 18 classes.
- Median per-instance duration error: 0.0% (more than half of all activity instances are measured at their exact annotated duration).

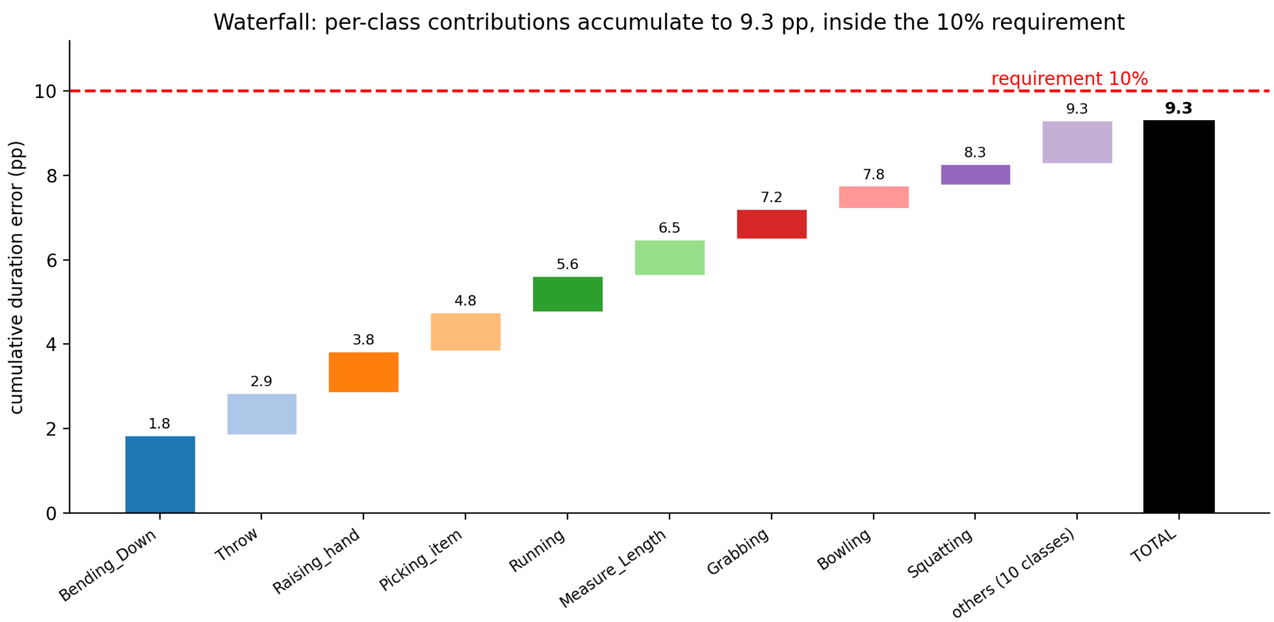

Figure 4. Waterfall decomposition of the overall duration error: each class contributes its mean error weighted by its share of instances (contribution = class mean error x n / total n); the contributions accumulate left to right to the total (black bar, 9.3 pp), which stays inside the 10% requirement (red dashed line). Cumulative values are annotated at each step; the ten smallest contributors (1.03 pp combined) are grouped as "others".

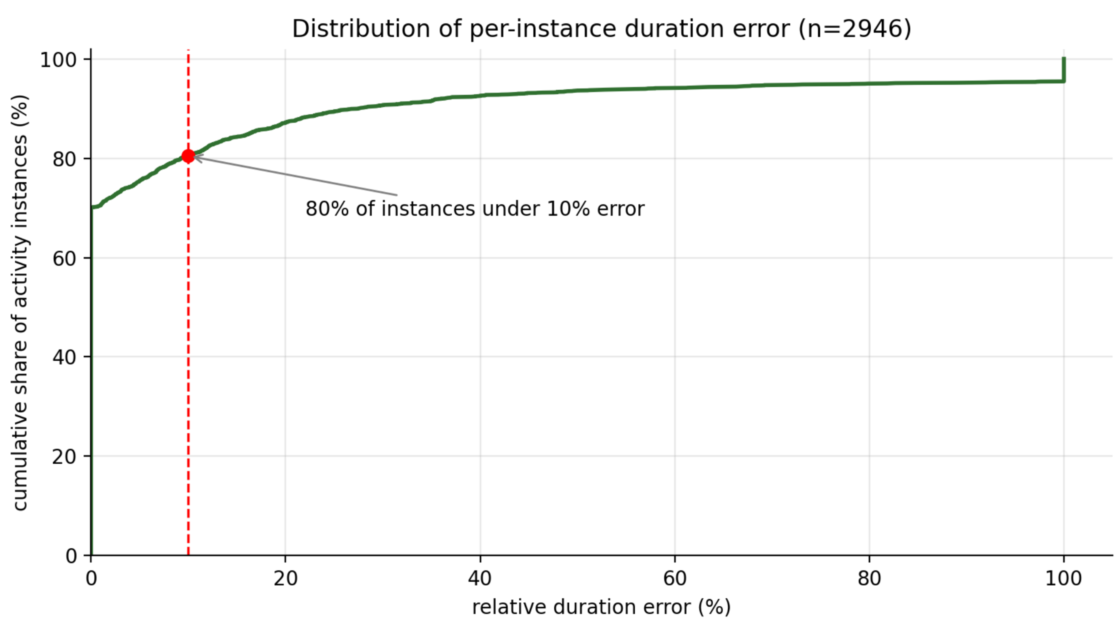

Figure 5. Cumulative distribution of the per-instance duration error: 70% of instances are measured with zero error, 80% within the 10% requirement. The long tail is concentrated in a small set of low-frequency classes (see waterfall).

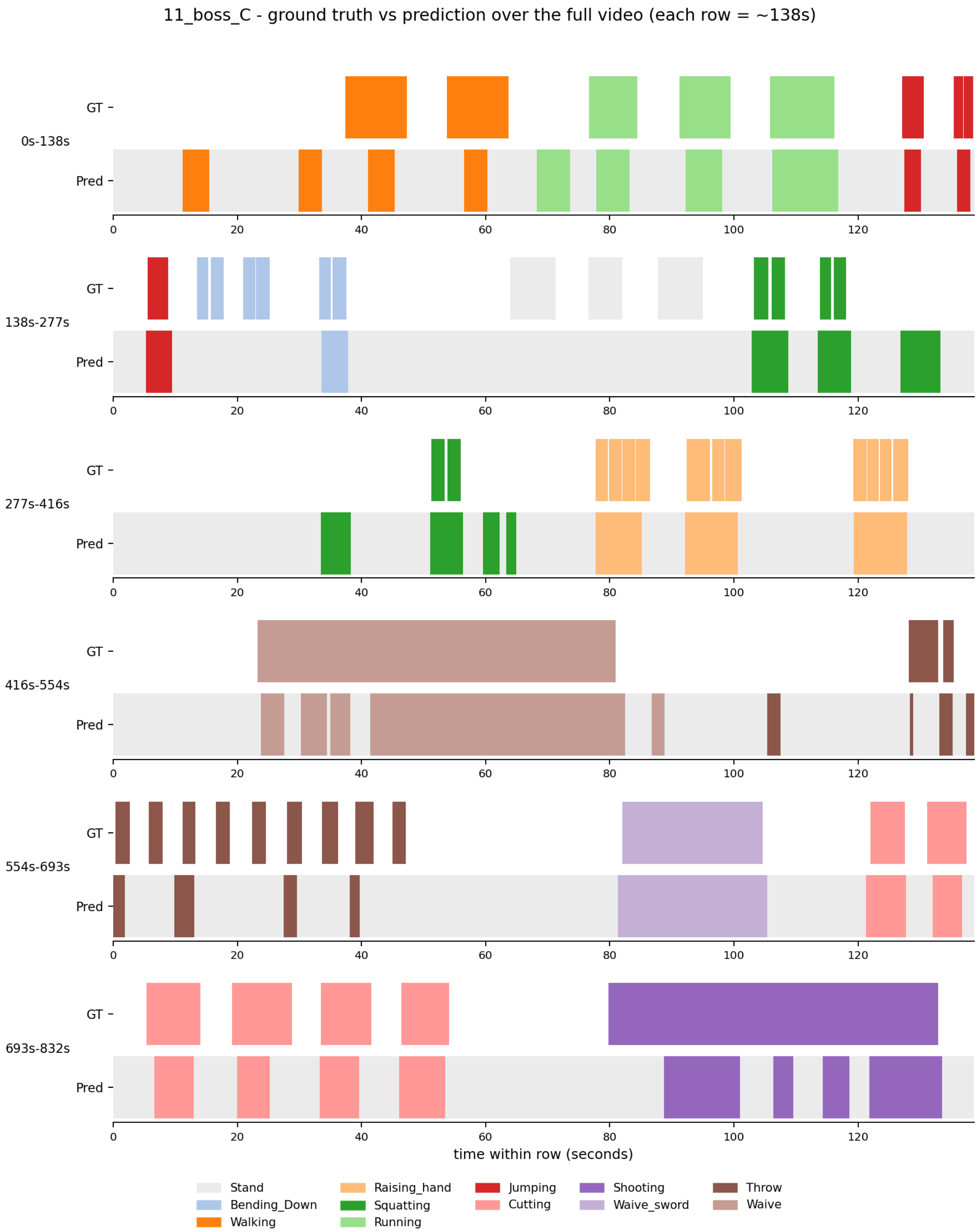

Figure 6. Example test video (11_boss_C, ~14 min): ground-truth label timeline vs model prediction, split into six ~138 s rows. White = unlabeled in the ground truth; light grey = Stand/background. Per-video timelines for all 63 test videos are archived under work_dirs/long_eval_18cls/eval_supp0.1/.

4. Reading the Results

All three downstream requirements are met: detection rate 96.80% against the 90% requirement, classification accuracy 99.70% against 85%, and activity duration error 9.3% against 10%.

Across every metric the errors share one structure: they are action-versus- background confusions, not action-versus-action. Once a clip is detected, its label is correct 99.70% of the time, and the confusion matrix (Figure 3) shows off-diagonal mass almost exclusively in the Stand row and column. Duration measurement follows the same pattern: half of all activity instances are measured at exactly their annotated duration (Figure 5), and the overall 9.3% error is dominated by a few classes — Bending_Down alone contributes 1.85 pp of the total, while the ten smallest contributors add only 1.03 pp combined (Figure 4). The three highest-frequency classes (Raising_hand, Jumping, Squatting — half of all instances) are each measured within 5% error. Qualitatively, predictions track the ground truth closely over full sessions (Figure 6), with residual misses concentrated in short repeated gestures.

Why the weak cases occur

The weak cases have identifiable causes rather than random model failure.

First, visual subtlety combined with scarce training data. Walking (50.0% clip-level detection) is slow, low-amplitude locomotion that is visually close to standing — and is treated as background by design in the deployed system. Measure_Length (64.1%) is a subtle controller gesture with almost no body movement. These are also the classes with the smallest training sample counts, so the model has both the hardest visual task and the least evidence to learn it from. If per-class targets are ever required, the straightforward remedy is additional training clips for these two classes.

Second, temporal granularity penalises short, repeated gestures. The inference window spans ~2.1 s, while gestures such as Throw last only ~1.5–2 s per repetition; windows that straddle an action boundary mix action and background frames and tend to be resolved as background. Each short repetition therefore loses a disproportionate share of its frames at the edges, which is why such classes score well at clip level (Throw: 110/110) yet lose duration in the long-video evaluation.

Third, the long-video evaluation runs on the raw session recordings, not on curated footage. Between task segments these videos contain unannotated stretches in which the trainee walks around, repositions, or idles between takes (including re-recorded attempts). These stretches carry no annotation and are therefore excluded from all three reported metrics, but they do produce a degree of false alarms: gesture-like movements during idle time account for part of the 4.94% clip-level false alarm rate, and they are visible in Figure 6 as occasional predicted activity inside unlabeled (white) regions. In deployment this effect can be suppressed by restricting analytics to the defined task phases of each session.